YUXUAN LI

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EDUCATION

University of California, Los Angeles *2019-2024*
Ph.D. in Mechanical Engineering. Major: Fluid Mechanics GPA: 4.0/4.0
Thesis Advisors: Jeff Eldredge and Adrienne Lavine

Shanghai Jiao Tong University *2015-2019*
B.S. in Aeronautics and Astronautics. GPA: 3.7/4.0

APPOINTMENTS

Simerics Inc., Novi, MI, USA
Project Engineer *2024-*

- **CFD Simulation and Analysis**: Develop high-fidelity 3D models using Simerics-MP+ to analyze complex fluid phenomena, including cavitation, aeration, and multiphase flow in rotating machinery.
- **Design Optimization and Validation**: Conduct transient simulations and parametric studies to evaluate component performance; validate numerical models against physical experimental data to ensure high-accuracy results.
- **Client Training and Technical Mentorship**: Deliver comprehensive training sessions on CFD best practices—from CAD-to-mesh translation to result analysis—and develop instructional documentation to facilitate client learning and methodology adoption.

JOURNAL PUBLICATIONS

1. Li, Y., Z. Wang, B. Yu, B. Zhang, and H. Liu. **"Gaussian models for late-time evolution of two-dimensional shock–light cylindrical bubble interaction."** *Shock Waves* 30, no. 2 (2020): 169-184.
2. Yuxuan Li, Jeff D. Eldredge, Adrienne S. Lavine, Timothy S. Fisher, and Bruce L. Drolen. **"A Conjugate Heat Transfer Model of Oscillating Heat Pipe Dynamics, Performance, and Dryout."** *International Journal of Heat and Mass Transfer* 227 (2024): 125530.
3. Yuxuan Li, Zachary Wong, Jeff D. Eldredge, Adrienne S. Lavine, Timothy S. Fisher, and Bruce L. Drolen. **"Effects of inclination angle on oscillating heat pipe thermal conductance and dryout."** *Under Review, ASME Journal of Heat and Mass Transfer*

CONFERENCES PAPERS

1. Yuxuan Li, Jeff D. Eldredge, Adrienne S. Lavine, Timothy S. Fisher, and Bruce L. Drolen. **"A data assimilation model of oscillating heat pipe dynamics and performance."** Joint 21st International Heat Pipe Conference and 15th International Heat Pipe Symposium, Melbourne, Australia (2023).
2. Yuxuan Li, Chengjie Wang. **"RANS-based simulation of KCS wave breaking characteristics."** Wageningen 2025: A Workshop on CFD in Ship Hydrodynamics, Wageningen, Netherlands (2025).

PRESENTATIONS

1. Li, Yuxuan, Jeff Eldredge, Adrienne Lavine, Timothy Fisher, and Bruce Drolen. **"Thermofluid modeling of an oscillating heat pipe."** The 16th Southern California Flow Physics Symposium, Los Angeles, USA (2022).
2. Yuxuan Li, Jeff Eldredge, Adrienne Lavine, Timothy Fisher, and Bruce Drolen. **"Thermofluid modeling of an oscillating heat pipe."** 75th Annual Meeting of the Division of Fluid Dynamics, Indianapolis, USA (2022).
3. Li, Yuxuan, Jeff Eldredge, Adrienne Lavine, Timothy Fisher, and Bruce Drolen. **"Thermofluid modeling of an oscillating heat pipe."** The 17th Southern California Flow Physics Symposium, San Diego, USA (2023).
4. Yuxuan Li, Zachary Wong, Jeff D. Eldredge, Timothy S. Fisher. **"Empirically Trained Models of Oscillating Heat Pipes for Improved Performance, Limits, and Control"** 2023 Spacecraft Thermal Control Workshop, Torrance, USA (2023)
5. Yuxuan Li, Jeff Eldredge, Adrienne Lavine, Timothy Fisher, and Bruce Drolen. **"Estimating thermofluid system parameters using a Markov chain Monte Carlo method, with an example of oscillating heat pipes."** 76th Annual Meeting of the Division of Fluid Dynamics, Washington D.C., USA (2023).

AWARDS

- UCLA Mechanical and Aerospace Engineering Department Fellowship *2019-2023*
- SJTU Outstanding Graduate *2019*
- UCLA-CSST Scholarship ($\sim$ 5000$) *2018*

TEACHING EXPERIENCE

University of California, Los Angeles

- Teaching Assistant for MAE150A *Intermediate Fluid Mechanics* (overall TA rating 7.85/9)

REVIEWER EXPERIENCE

ASME Journal of Heat and Mass Transfer
International Journal of Thermal Sciences
SAE WCX 2026